HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed low-level **C** device drivers for hardware interfaces including **GPIOs** and **interrupt controllers** on ARM-based processors, collaborating directly with hardware designers.
- Wrote embedded **C** code for power management features, ensuring efficient operation across various hardware states for a semiconductor client.
- Engineered Python scripts for automating data extraction and analysis from embedded system logs, reducing manual effort by approximately **40%**.
- Implemented tooling in **Python** to streamline the debugging process for embedded software, significantly improving diagnostic efficiency for the team.
- Debugged complex hardware/software interaction issues using **Lauterbach Trace32** and JTAG, resolving critical bugs that impacted device stability.
- Integrated new hardware peripherals by developing and testing low-level device drivers, ensuring seamless communication within the SoC architecture.
- Managed software configuration updates and coordinated user acceptance testing for enterprise systems, ensuring smooth transitions during change control processes.

SKILLS

Languages – C, Python, SQL, Java

Embedded Systems – Device Drivers, Hardware/Software Interface, ARM, DSP, RISC-V, Power Management, GPIO, Interrupts, JTAG, Lauterbach Trace32

Tools & Technologies – Docker, Linux, Git, Jenkins, Jira, Splunk, Dynatrace

Databases – PostgreSQL, MySQL, Oracle

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

PROJECTS

Embedded C Driver Development for SoC Peripherals

- Developed and tested low-level **C** drivers for digital clock controllers and voltage regulators on an ARM Cortex-M processor.
- Collaborated with hardware architects to define and implement precise hardware/software interfaces, reducing signal integrity issues by **15%**.
- Integrated drivers into the main firmware build using **Makefiles** and performed initial hardware bring-up tests.

Python Automation for Embedded System Debugging

- Created a suite of **Python** scripts to automate log parsing and error reporting from embedded devices, cutting analysis time by an average of **3 hours** per debugging session.
- Developed a small CLI tool in **Python** to trigger specific test sequences on target hardware via a JTAG interface, aiding in regression testing.
- Documented the automation framework and provided training to junior engineers on its usage.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)